

UNIVERSITY OF BRISTOL

Faculty of Science

**MOCK EXAMINATION 4 — SOLUTIONS AND
MARK SCHEME****Core Physics I — Classical, Quantum and Thermal Physics**

Module Code: PHYS10012

Total Marks: 100

Marking Notes:

- Mark allocations are shown in **blue** at each step.
 - Final answers are highlighted in green boxes.
 - “*(From formula sheet)*” indicates the equation is on the provided formula sheet.
 - “*(Recalled — not on formula sheet)*” indicates the equation must be known from memory.
 - Award partial marks for correct method even if the final answer is wrong.
 - Accept equivalent correct forms of answers.
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Section A: Oscillations and Waves — Answers

1. **B** — The period of a simple pendulum is $T = 2\pi\sqrt{L/g}$ (*From formula sheet*), so $T \propto \sqrt{L}$. For the period to double, we need $2T \propto \sqrt{L'}$, giving $L' = 4L$.
2. **B** — The total energy of a simple harmonic oscillator is $E = \frac{1}{2}kA^2$ (*Recalled — not on formula sheet*). Thus $E = \frac{1}{2}(200)(0.1)^2 = \frac{1}{2}(200)(0.01) = 1.0$ J.
3. **B** — Critical damping is the boundary between underdamped ($\gamma < 2\omega_0$) and overdamped ($\gamma > 2\omega_0$) motion. It occurs when $\gamma = 2\omega_0$ (*Recalled — not on formula sheet*).
4. **C** — The average power absorbed by a driven oscillator is proportional to the imaginary part of the response, which peaks at $\omega = \omega_0$ (not at the amplitude resonance frequency $\omega_{\max} = \sqrt{\omega_0^2 - \gamma^2/2}$). Maximum power transfer occurs at $\omega = \omega_0$.
5. **C** — The wave speed on a string is $v = \sqrt{T/\mu}$ (*From formula sheet*). If the tension is quadrupled ($T \rightarrow 4T$), the new speed is $v' = \sqrt{4T/\mu} = 2\sqrt{T/\mu} = 2v$.
6. **A** — The amplitude transmission coefficient is $t = 2Z_1/(Z_1 + Z_2)$ (*From formula sheet*). With $Z_2 = 3Z_1$: $t = 2Z_1/(Z_1 + 3Z_1) = 2Z_1/(4Z_1) = 1/2$.
7. **C** — From the decibel formula $\beta = 10 \log_{10}(I/I_0)$ (*From formula sheet*): $60 = 10 \log_{10}(I/10^{-12})$, so $\log_{10}(I/10^{-12}) = 6$, giving $I/10^{-12} = 10^6$, hence $I = 10^{-6}$ W/m².
8. **C** — The wall acts as a moving observer receiving frequency $f_1 = f_0(v+v_w)/v$. The wall then re-emits at this frequency as a moving source approaching the stationary source, so the frequency heard is $f_2 = f_1 \cdot v/(v-v_w) = f_0(v+v_w)/(v-v_w) = 500 \times 350/330 = 530.3$ Hz ≈ 530 Hz.
9. **B** — Open-open pipe: $f_n = nv/(2L)$, so the 2nd harmonic is $f_2 = 2v/(2L) = v/L$. Open-closed pipe: $f_n = nv/(4L)$ with $n = 1, 3, 5, \dots$, so the fundamental is $f_1 = v/(4L)$. The ratio is $(v/L)/(v/(4L)) = 4$.
10. **B** — Higher tension gives a higher frequency via $v = \sqrt{T/\mu}$ and $f \propto v$ (for a given string length). Since string B has higher tension, $f_B > 440$ Hz. The beat frequency is $|f_B - f_A| = 3$ Hz, so $f_B = 443$ Hz.

Section B: Quantum Physics — Answers

1. **C** — $|\psi\rangle\langle\phi|$ is an outer product, which is a linear operator. It maps any ket $|\chi\rangle$ to $|\psi\rangle\langle\phi|\chi\rangle$, which is a ket times a scalar, i.e. another ket.
2. **C** — A global phase factor $e^{i\theta}$ does not affect any measurement probabilities: $|\langle a|e^{i\theta}\psi\rangle|^2 = |e^{i\theta}|^2|\langle a|\psi\rangle|^2 = |\langle a|\psi\rangle|^2$. The two states are physically indistinguishable.
3. **C** — Using a complete orthonormal basis $\{|j\rangle\}$: $\text{Tr}(|a\rangle\langle b|) = \sum_j \langle j|a\rangle\langle b|j\rangle = \sum_j \langle j|a\rangle\langle b|j\rangle = \sum_j \langle b|j\rangle\langle j|a\rangle = \langle b|(\sum_j |j\rangle\langle j|)|a\rangle = \langle b|a\rangle$.
4. **B** — Using the spin operators from the formula sheet, one can verify $[S_x, S_y] = S_x S_y - S_y S_x = i\hbar S_z$. This is the fundamental angular momentum commutation relation.
5. **C** — By the Born rule (*Recalled — not on formula sheet*), $P(\downarrow_z) = |\langle\downarrow_z|\psi\rangle|^2 = \left|\sqrt{2/3}\right|^2 = 2/3$.
6. **C** — After measuring $S_z = +\hbar/2$, the state is $|\uparrow_z\rangle$. We can write $|\uparrow_z\rangle = \frac{1}{\sqrt{2}}|\uparrow_y\rangle + \frac{1}{\sqrt{2}}|\downarrow_y\rangle$ (*From formula sheet*). Therefore $P(\uparrow_y) = |1/\sqrt{2}|^2 = 1/2$.
7. **B** — A stationary state is an eigenstate of the Hamiltonian. Under time evolution it acquires only a global phase: $|E_n\rangle \rightarrow e^{-iE_n t/\hbar}|E_n\rangle$. Since the global phase cancels in all probability calculations $|\langle a|\psi(t)\rangle|^2$, the measurement probabilities in *any* basis are time-independent.
8. **C** — Each spin- $\frac{1}{2}$ particle has a 2-dimensional state space. The composite space is the tensor product: $\dim = 2 \times 2 = 4$.
9. **B** — The Pauli exclusion principle states that no two identical fermions can occupy the same quantum state. This follows from the requirement that multi-fermion states must be antisymmetric under particle exchange, which is a consequence of the spin-statistics theorem for half-integer spin particles.
10. **C** — Beta-minus decay involves the transformation of a down quark into an up quark ($d \rightarrow u + e^- + \bar{\nu}_e$), which is mediated by the exchange of a W^- boson. This is a weak interaction process.

Section C: Long Answer Solutions

Question 1.

[15 marks]

Newton's Laws, Friction, and Circular Motion

- (a) (i) The maximum static friction force is (*Recalled — not on formula sheet*):

$$f_{s,\max} = \mu_s m_1 g = 0.4 \times 5 \times 9.81 = 19.62 \text{ N}$$

[1] for the correct formula;

[1] for the numerical answer.

$$f_{s,\max} = 19.6 \text{ N. The block will not move for any applied force } F \leq 19.6 \text{ N.}$$

- (ii) Since $F = 25 \text{ N} > 19.6 \text{ N}$, the block moves and kinetic friction applies (*Recalled — not on formula sheet*):

$$F - \mu_k m_1 g = m_1 a$$

$$25 - 0.3 \times 5 \times 9.81 = 5a$$

$$25 - 14.72 = 5a \implies a = \frac{10.28}{5} = 2.06 \text{ m/s}^2$$

[1] for the equation of motion;

[1] for the numerical answer.

$$a = 2.06 \text{ m/s}^2$$

- (b) First, determine the direction of acceleration. Compare the gravitational components:

$$m_1 g \sin 30^\circ = 5 \times 9.81 \times 0.5 = 24.53 \text{ N (component pulling block down the incline)}$$

$$m_2 g = 3 \times 9.81 = 29.43 \text{ N (weight of hanging mass)}$$

Since $m_2 g > m_1 g \sin \theta$, the hanging mass descends and the block moves up the incline.

[1]

For the block on the incline (taking up the incline as positive):

$$T - m_1 g \sin 30^\circ = m_1 a \implies T - 24.53 = 5a \quad (1)$$

For the hanging mass (taking downward as positive):

$$m_2 g - T = m_2 a \implies 29.43 - T = 3a \quad (2)$$

[1] for both correct equations of motion.

Adding (1) and (2):

$$29.43 - 24.53 = (5 + 3)a \implies 4.90 = 8a \implies a = 0.613 \text{ m/s}^2$$

[1]

Substituting back into (2):

$$T = 29.43 - 3 \times 0.613 = 29.43 - 1.84 = 27.59 \text{ N}$$

[1]

$$a = 0.61 \text{ m/s}^2 \text{ (block accelerates up the incline); } T = 27.6 \text{ N.}$$

- (c) On a banked curve with no friction, the only forces on the car are its weight Mg (downward) and the normal force N (perpendicular to the road surface). Resolving forces: [1]

Vertical: $N \cos \alpha = Mg$

Horizontal (centripetal): $N \sin \alpha = \frac{Mv^2}{r}$

Dividing: [1]

$$\tan \alpha = \frac{v^2}{rg}$$

$$v = \sqrt{rg \tan \alpha} = \sqrt{80 \times 9.81 \times \tan 20^\circ} = \sqrt{80 \times 9.81 \times 0.3640} = \sqrt{285.7} = 16.9 \text{ m/s}$$
 [1]

Design speed $v = 16.9 \text{ m/s}$ ($\approx 60.8 \text{ km/h}$)

- (d) For the conical pendulum, resolve forces on the mass: [1]

Vertical: $T \cos \theta = mg$

$$T = \frac{mg}{\cos \theta} = \frac{0.5 \times 9.81}{\cos 25^\circ} = \frac{4.905}{0.9063} = 5.41 \text{ N}$$

[1]

Radius of circular orbit:

$$r = L \sin \theta = 1.0 \times \sin 25^\circ = 0.4226 \text{ m} \approx 0.423 \text{ m}$$

[1]

Horizontal (centripetal): $T \sin \theta = m\omega^2 r$

$$\omega^2 = \frac{g}{L \cos \theta} = \frac{9.81}{1.0 \times 0.9063} = 10.82 \text{ rad}^2/\text{s}^2$$

$$\omega = 3.290 \text{ rad/s}, \quad T_{\text{period}} = \frac{2\pi}{\omega} = \frac{2\pi}{3.290} = 1.91 \text{ s}$$

[1]

$$T_{\text{tension}} = 5.41 \text{ N}, \quad r = 0.423 \text{ m}, \quad T_{\text{period}} = 1.91 \text{ s}$$

Question 2.

[15 marks]

Maxwell–Boltzmann, Lennard-Jones, and Crystal Structure

- (a) The three characteristic speeds from the Maxwell–Boltzmann distribution are (*From formula sheet*):

$$c_{\text{mp}} = \sqrt{\frac{2k_B T}{m}} [1]$$

$$c_{\text{avg}} = \sqrt{\frac{8k_B T}{\pi m}} [1]$$

$$c_{\text{rms}} = \sqrt{\frac{3k_B T}{m}} [1]$$

- (b) The molecular mass of N_2 is:

$$m = \frac{M}{N_A} = \frac{28 \times 10^{-3}}{6.022 \times 10^{23}} = 4.65 \times 10^{-26} \text{ kg}$$

[1]

$$c_{\text{mp}} = \sqrt{\frac{2 \times 1.381 \times 10^{-23} \times 300}{4.65 \times 10^{-26}}} = \sqrt{\frac{8.286 \times 10^{-21}}{4.65 \times 10^{-26}}} = \sqrt{1.782 \times 10^5} = 422 \text{ m/s}$$

[1]

$$c_{\text{avg}} = \sqrt{\frac{8 \times 1.381 \times 10^{-23} \times 300}{\pi \times 4.65 \times 10^{-26}}} = \sqrt{\frac{3.314 \times 10^{-20}}{1.461 \times 10^{-25}}} = \sqrt{2.268 \times 10^5} = 476 \text{ m/s}$$

[1]

$$c_{\text{rms}} = \sqrt{\frac{3 \times 1.381 \times 10^{-23} \times 300}{4.65 \times 10^{-26}}} = \sqrt{\frac{1.243 \times 10^{-20}}{4.65 \times 10^{-26}}} = \sqrt{2.673 \times 10^5} = 517 \text{ m/s}$$

[1]

$c_{\text{mp}} = 422 \text{ m/s}, \quad c_{\text{avg}} = 476 \text{ m/s}, \quad c_{\text{rms}} = 517 \text{ m/s}.$
 Since $422 < 476 < 517$, we confirm $c_{\text{mp}} < c_{\text{avg}} < c_{\text{rms}}$.

- (c) The Lennard-Jones potential is (*From formula sheet*):

$$V(r) = 4\epsilon \left[\left(\frac{a}{r} \right)^{12} - \left(\frac{a}{r} \right)^6 \right]$$

Differentiating:

$$\frac{dV}{dr} = 4\epsilon \left[-12 \frac{a^{12}}{r^{13}} + 6 \frac{a^6}{r^7} \right] = 0$$

[1]

$$12 \frac{a^{12}}{r^{13}} = 6 \frac{a^6}{r^7} \implies 2a^6 = r^6 \implies r_{\text{eq}} = 2^{1/6} a$$

[1]

Substituting $r_{\text{eq}} = 2^{1/6} a$ into $V(r)$:

$$V(r_{\text{eq}}) = 4\varepsilon \left[\left(\frac{a}{2^{1/6}a} \right)^{12} - \left(\frac{a}{2^{1/6}a} \right)^6 \right] = 4\varepsilon [2^{-2} - 2^{-1}] = 4\varepsilon \left[\frac{1}{4} - \frac{1}{2} \right] = -\varepsilon$$

[1]

$$r_{\text{eq}} = 2^{1/6} a \approx 1.122 a. \quad \text{The binding energy is } |V(r_{\text{eq}})| = \varepsilon.$$

[1]

(d) A BCC unit cell contains 2 atoms (*Recalled — not on formula sheet*).

[1]

Mass of one iron atom:

$$m_{\text{atom}} = 55.85 \times 1.661 \times 10^{-27} = 9.277 \times 10^{-26} \text{ kg}$$

[1]

Volume of the unit cell:

$$V_{\text{cell}} = a_{\text{latt}}^3 = (2.87 \times 10^{-10})^3 = 2.364 \times 10^{-29} \text{ m}^3$$

[1]

Density:

$$\rho = \frac{2 m_{\text{atom}}}{a_{\text{latt}}^3} = \frac{2 \times 9.277 \times 10^{-26}}{2.364 \times 10^{-29}} = \frac{1.855 \times 10^{-25}}{2.364 \times 10^{-29}} = 7850 \text{ kg/m}^3$$

[1]

$$\rho \approx 7850 \text{ kg/m}^3 \text{ (the known density of iron is } \approx 7874 \text{ kg/m}^3\text{).}$$

Question 3.

[15 marks]

SHM, Superposition, and Organ Pipes(a) The angular frequency is (*From formula sheet*):

$$\omega_0 = \sqrt{\frac{k}{m}} = \sqrt{\frac{160}{0.4}} = \sqrt{400} = 20 \text{ rad/s}$$

[1]

Released from rest at $x = 0.08 \text{ m}$, so $A = 0.08 \text{ m}$ and $\phi = 0$ (*From formula sheet*):

$$x(t) = 0.08 \cos(20t) \text{ m}$$

Maximum velocity (*Recalled — not on formula sheet*):

$$v_{\max} = A\omega_0 = 0.08 \times 20 = 1.6 \text{ m/s}$$

[1]

Maximum acceleration (*Recalled — not on formula sheet*):

$$a_{\max} = A\omega_0^2 = 0.08 \times 400 = 32 \text{ m/s}^2$$

[1]

$$x(t) = 0.08 \cos(20t) \text{ m}, \quad v_{\max} = 1.6 \text{ m/s}, \quad a_{\max} = 32 \text{ m/s}^2$$

(b) Using $x(t) = A \cos(\omega_0 t + \phi)$ (*From formula sheet*), with $\omega_0 = 20 \text{ rad/s}$:At $t = 0$: $x(0) = A \cos \phi = 0.06 \text{ m}$ $v(0) = -A\omega_0 \sin \phi = 1.20 \text{ m/s}$, so $A \sin \phi = -v_0/\omega_0 = -1.20/20 = -0.06 \text{ m}$. [1]

Amplitude:

$$A = \sqrt{x_0^2 + (v_0/\omega_0)^2} = \sqrt{0.06^2 + 0.06^2} = \sqrt{0.0072} = 0.0849 \text{ m} \approx 8.49 \text{ cm}$$

[1]

Phase:

$$\tan \phi = \frac{-v_0/\omega_0}{x_0} = \frac{-0.06}{0.06} = -1$$

Since $\cos \phi > 0$ and $\sin \phi < 0$, we are in the fourth quadrant: $\phi = -\pi/4$. [1]

$$A = 0.0849 \text{ m} \approx 8.49 \text{ cm}, \quad \phi = -\pi/4, \quad x(t) = 0.0849 \cos(20t - \pi/4) \text{ m}.$$

[1]

(c) The two waves travelling in the $+x$ direction are:

$$y_1 = A \sin(kx - \omega t), \quad y_2 = A \sin(kx - \omega t + \delta)$$

where $\delta = \pi/3$.

Wavenumber and angular frequency:

[1]

$$k = \frac{2\pi f}{v} = \frac{2\pi \times 200}{100} = 4\pi \text{ rad/m}, \quad \omega = 2\pi f = 400\pi \text{ rad/s}$$

Resultant amplitude (using the superposition formula for two waves of equal amplitude) (*Recalled — not on formula sheet*):

$$A_R = 2A \cos\left(\frac{\delta}{2}\right) = 2 \times 3 \times \cos\left(\frac{\pi}{6}\right) = 6 \times \frac{\sqrt{3}}{2} = 3\sqrt{3} \approx 5.20 \text{ mm}$$

[1]

The resultant wave has a phase shifted by $\delta/2$ relative to y_1 :

$$y = A_R \sin\left(kx - \omega t + \frac{\delta}{2}\right) = 3\sqrt{3} \sin\left(4\pi x - 400\pi t + \frac{\pi}{6}\right) \text{ mm}$$

[1] for the resultant amplitude; [1] for the complete wave equation.

$$A_R = 3\sqrt{3} \approx 5.20 \text{ mm.} \quad y = 3\sqrt{3} \sin(4\pi x - 400\pi t + \pi/6) \text{ mm.}$$

- (d) (i) Open-open pipe (*From formula sheet*): $f_n = \frac{nv}{2L}$, $n = 1, 2, 3, \dots$

$$f_1 = \frac{340}{2 \times 1.7} = 100 \text{ Hz}, \quad f_2 = 200 \text{ Hz}, \quad f_3 = 300 \text{ Hz}, \quad f_4 = 400 \text{ Hz}$$

[1] for the formula; [1] for all four frequencies.

$$f_1 = 100 \text{ Hz}, f_2 = 200 \text{ Hz}, f_3 = 300 \text{ Hz}, f_4 = 400 \text{ Hz.}$$

- (ii) Open-closed pipe (*From formula sheet*): $f_n = \frac{nv}{4L}$, $n = 1, 3, 5, \dots$ (odd harmonics only)

$$f_1 = \frac{340}{4 \times 1.7} = 50 \text{ Hz}, \quad f_3 = 150 \text{ Hz}, \quad f_5 = 250 \text{ Hz}$$

[1] for recognising odd harmonics only; [1] for the frequencies.

$$f_1 = 50 \text{ Hz}, f_3 = 150 \text{ Hz}, f_5 = 250 \text{ Hz.}$$

Question 4.

[15 marks]

Time Evolution and Multi-Particle States

- (a) The energy eigenstates of $H = \omega_0 S_z$ are $|\uparrow\rangle$ with eigenvalue $E_{\uparrow} = +\omega_0 \hbar/2$ and $|\downarrow\rangle$ with eigenvalue $E_{\downarrow} = -\omega_0 \hbar/2$. [1]

At $t = 0$, with $\alpha = \pi/6$:

$$|\psi(0)\rangle = \cos\left(\frac{\pi}{6}\right) |\uparrow\rangle + \sin\left(\frac{\pi}{6}\right) |\downarrow\rangle = \frac{\sqrt{3}}{2} |\uparrow\rangle + \frac{1}{2} |\downarrow\rangle$$

Using time evolution (*From formula sheet*): $|\psi(t)\rangle = \sum_n c_n e^{-iE_n t/\hbar} |E_n\rangle$:

$$|\psi(t)\rangle = \frac{\sqrt{3}}{2} e^{-i\omega_0 t/2} |\uparrow\rangle + \frac{1}{2} e^{+i\omega_0 t/2} |\downarrow\rangle$$

[1]

Probabilities:

$$P(\uparrow_z, t) = \left| \frac{\sqrt{3}}{2} e^{-i\omega_0 t/2} \right|^2 = \frac{3}{4}$$

$$P(\downarrow_z, t) = \left| \frac{1}{2} e^{+i\omega_0 t/2} \right|^2 = \frac{1}{4}$$

Both probabilities are **time-independent**. [1]

This is expected: the state was written in the energy eigenbasis, so probabilities of energy (or equivalently S_z) measurements are constant. [1]

$$|\psi(t)\rangle = \frac{\sqrt{3}}{2} e^{-i\omega_0 t/2} |\uparrow\rangle + \frac{1}{2} e^{+i\omega_0 t/2} |\downarrow\rangle.$$

$P(\uparrow_z) = 3/4$ and $P(\downarrow_z) = 1/4$, both constant in time.

- (b) The spin operator $S_x = \frac{\hbar}{2} (|\uparrow\rangle\langle\downarrow| + |\downarrow\rangle\langle\uparrow|)$ (*From formula sheet*).

$$\langle S_x \rangle(t) = \langle \psi(t) | S_x | \psi(t) \rangle.$$

[1]

Computing:

$$\begin{aligned} \langle S_x \rangle &= \frac{\hbar}{2} [\langle \psi | \uparrow \rangle \langle \downarrow | \psi \rangle + \langle \psi | \downarrow \rangle \langle \uparrow | \psi \rangle] \\ &= \frac{\hbar}{2} [\langle \psi | \uparrow \rangle \langle \downarrow | \psi \rangle + \langle \psi | \downarrow \rangle \langle \uparrow | \psi \rangle] \end{aligned}$$

Now:

$$\langle \uparrow | \psi(t) \rangle = \frac{\sqrt{3}}{2} e^{-i\omega_0 t/2}, \quad \langle \downarrow | \psi(t) \rangle = \frac{1}{2} e^{+i\omega_0 t/2}$$

So:

$$\begin{aligned}
 \langle S_x \rangle &= \frac{\hbar}{2} \left[\frac{\sqrt{3}}{2} e^{+i\omega_0 t/2} \cdot \frac{1}{2} e^{+i\omega_0 t/2} + \frac{1}{2} e^{-i\omega_0 t/2} \cdot \frac{\sqrt{3}}{2} e^{-i\omega_0 t/2} \right] \\
 &= \frac{\hbar}{2} \left[\frac{\sqrt{3}}{4} e^{+i\omega_0 t} + \frac{\sqrt{3}}{4} e^{-i\omega_0 t} \right] \\
 &= \frac{\hbar}{2} \cdot \frac{\sqrt{3}}{2} \cos(\omega_0 t) = \frac{\sqrt{3}\hbar}{4} \cos(\omega_0 t)
 \end{aligned}$$

[2]

$\langle S_x \rangle = 0$ when $\cos(\omega_0 t) = 0$, i.e.:

$$\omega_0 t = \frac{\pi}{2} + n\pi \quad (n = 0, 1, 2, \dots) \implies t = \frac{(2n+1)\pi}{2\omega_0}$$

[1]

$$\langle S_x \rangle(t) = \frac{\sqrt{3}\hbar}{4} \cos(\omega_0 t). \quad \langle S_x \rangle = 0 \text{ at } t = \frac{(2n+1)\pi}{2\omega_0}, n = 0, 1, 2, \dots$$

- (c) For two distinguishable spin- $\frac{1}{2}$ particles, the composite state space is spanned by the tensor product of the individual bases. A complete orthonormal basis is:

[1]

$$|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle$$

where the first arrow refers to particle 1 and the second to particle 2.

There are $2 \times 2 = 4$ basis states.

[1]

An example of an entangled state (one that cannot be written as a product $|a\rangle_1 \otimes |b\rangle_2$):

[1]

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle) \quad (\text{the singlet state})$$

Basis: $\{|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle\}$, with 4 states total.

Example entangled state: $\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$.

- (d) **Two identical bosons** in states $\{|a\rangle, |b\rangle, |c\rangle\}$:

Bosonic states must be symmetric under particle exchange. The allowed states are:

[1]

Both in the same state (3 states):

$$|a\rangle|a\rangle, |b\rangle|b\rangle, |c\rangle|c\rangle$$

One in each of two different states (3 states):

$$\frac{1}{\sqrt{2}}(|a\rangle|b\rangle + |b\rangle|a\rangle), \quad \frac{1}{\sqrt{2}}(|a\rangle|c\rangle + |c\rangle|a\rangle), \quad \frac{1}{\sqrt{2}}(|b\rangle|c\rangle + |c\rangle|b\rangle)$$

Total: **6** two-boson states.

[1]

This agrees with the combinatorial formula $\binom{n+k-1}{k} = \binom{3+2-1}{2} = \binom{4}{2} = 6$.

Two identical fermions in states $\{|a\rangle, |b\rangle, |c\rangle\}$:

Fermionic states must be antisymmetric under exchange. Both particles cannot be in the same state (Pauli exclusion). The allowed states are:

[1]

$$\frac{1}{\sqrt{2}} (|a\rangle |b\rangle - |b\rangle |a\rangle), \quad \frac{1}{\sqrt{2}} (|a\rangle |c\rangle - |c\rangle |a\rangle), \quad \frac{1}{\sqrt{2}} (|b\rangle |c\rangle - |c\rangle |b\rangle)$$

Total: **3** two-fermion states.

[1]

This agrees with $\binom{n}{k} = \binom{3}{2} = 3$.

Bosons: 6 states (3 with both particles in same state + 3 symmetric combinations).

Fermions: 3 states (antisymmetric combinations only; no double occupancy).